23A31A0593

Course Code:23CS402T

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PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)

II B.Tech II Semester Regular Examinations, April - 2025

SOFTWARE ENGINEERING

(Common to CSE and IT)

Time: 3 hours **Max. Marks: 70**

Note:

i. *Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.*

ii. *In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.*

Q. No.		Questions	BTL	CO	Marks
1.	a)	What is the Waterfall model in software engineering?	K1	CO1	2M
	b)	Define Agile methodology. List any two agile models.	K2	CO1	2M
	c)	List any two responsibilities of a software project manager.	K2	CO2	2M
	d)	What are empirical estimation techniques?	K2	CO2	2M
	e)	Define cohesion. Mention its types.	K1	CO3	2M
	f)	List the types of user interfaces.	K2	CO3	2M
	g)	What is black box testing?	K2	CO4	2M
	h)	Mention two objectives of code review.	K1	CO4	2M
	i)	What is software reuse?	K1	CO5	2M
	j)	Define software reverse engineering.	K1	CO5	2M
		UNIT – I			
2.	a)	Explain the differences between software products and software services.	K2	CO1	5M
	b)	Describe the Spiral model with its advantages and limitations.	K2	CO1	5M
		OR			
3.	a)	Explain the Rapid Application Development model. How does it improve productivity?	K2	CO1	5M
	b)	Illustrate the role of computer systems engineering in software projects.	K3	CO1	5M
		UNIT – II			
4.	a)	Illustrate the working of the basic COCOMO model for project cost estimation.	K4	CO2	5M
	b)	Discuss the algebraic and axiomatic specification techniques in formal system specification.	K2	CO2	5M
		OR			
5.	a)	Outline the characteristics of a good Software Requirements Specification (SRS) document.	K2	CO2	5M
	b)	Compute the Function Point (FP) metric with a suitable example.	K3	CO2	5M
		UNIT – III			
6.	a)	Analyze the difference between cohesion and coupling with examples.	K4	CO3	5M
	b)	Explain the core practices and benefits of Extreme Programming (XP).	K2	CO3	5M

		OR			
7.	a)	Summarize the steps involved in structured analysis during function-oriented design.	K3	CO3	5M
	b)	Explain any four types of user interfaces with examples.	K2	CO3	5M
		UNIT – IV			
8.	a)	Differentiate between white-box and black-box testing strategies.	K2	CO4	5M
	b)	Describe two major types of reviews conducted during code review.	K2	CO4	5M
		OR			
9.	a)	Discuss the various levels in SEI Capability Maturity Model (CMM).	K3	CO4	5M
	b)	Describe the importance and process of integration testing.	K2	CO4	5M
		UNIT – V			
10.	a)	Explain the architecture of a CASE environment and its components.	K2	CO5	5M
	b)	Summarize Lehman's laws of software evolution.	K2	CO5	5M
		OR			
11.	a)	Define software reuse. Describe the organizational steps to implement reuse.	K2	CO5	5M
	b)	Outline the process and benefits of software reverse engineering.	K2	CO5	5M